

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL3_Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I _Nandhini sri V_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.
2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.
3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

During seasonal sales such as Diwali, Black Friday, or festival offers, an e-commerce website may receive a very large number of users within a short time.

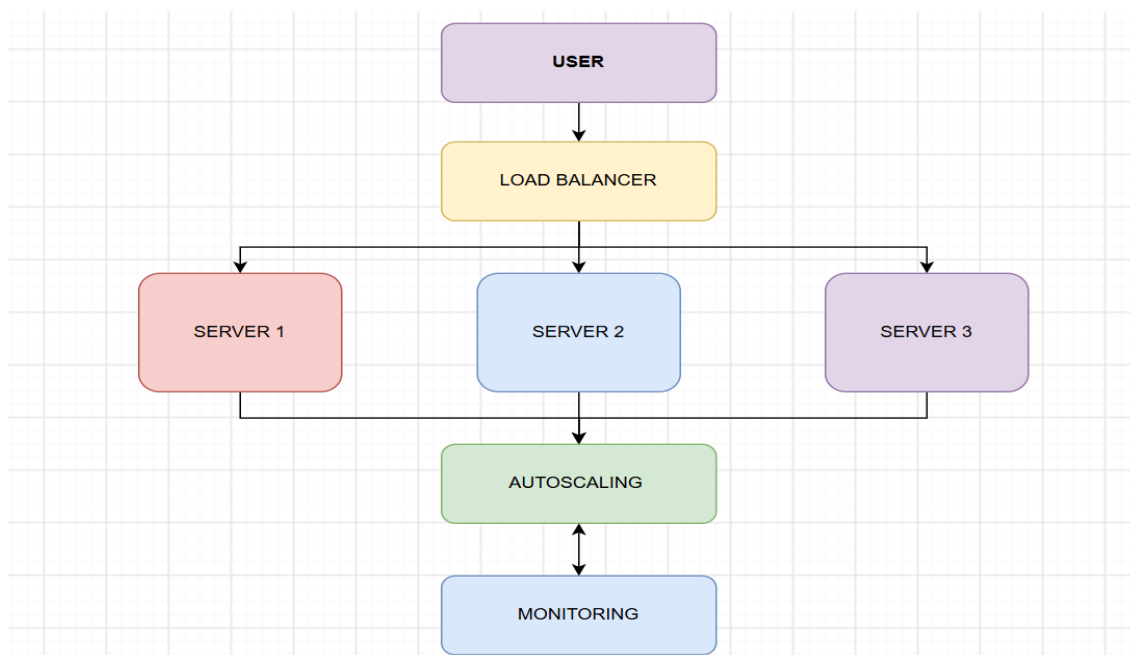
auto-scaling

- **Auto-scaling** automatically increases or decreases the number of cloud servers based on demand.
- During a traffic spike, new instances are added to handle additional requests.
- When traffic falls, unnecessary instances are removed.
- This avoids both **server overload** and **unnecessary cloud costs**.

Role of load balancing

A load balancer distributes incoming customer requests across multiple servers instead of sending everything to one server.

For example:



Customers → Load Balancer → Server 1 / Server 2 / Server 3 / Server 4

If one server becomes unavailable, the load balancer can redirect users to healthy servers.

Reducing downtime and improving user experience

This combination helps:

- Prevent website crashes during peak traffic.
- Reduce page loading time.
- Keep checkout and payment services available.
- Maintain service even when one server fails.
- Provide a smoother shopping experience.

Role of monitoring tools

Monitoring tools continuously track:

- CPU and memory usage
- Number of users/requests
- Response time
- Server health
- Error rates

When traffic crosses a predefined limit, monitoring data can trigger auto-scaling.

Example:

Consider Amazon during a major sale. A sudden increase in customers searching for products and placing orders can cause heavy traffic. Auto-scaling adds computing resources, while load balancing distributes requests among available servers. Monitoring helps the system detect the increase and respond quickly.

Thus, load balancing distributes the workload while auto-scaling adjusts the required resources. Together, they improve availability, reduce downtime, and provide a better shopping experience.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

ANSWER:

An organization may have terabytes of backup data containing business records, customer information, and application data. Storing all of this on physical storage systems can be expensive and difficult to maintain.

Durability and data redundancy

Cloud storage providers maintain multiple copies of data across different storage devices or locations.

- If one storage device fails, another copy can be used.
- Data can be replicated across different availability zones or locations.
- This provides high durability and fault tolerance.
- Regular backups and versioning can also help recover accidentally deleted files.

Encryption and access control:

Sensitive backup data can be protected using:

Encryption:

- Data is encrypted while stored (**at rest**).
- Data can also be encrypted while being transferred (**in transit**).
- Even if unauthorized users obtain the storage files, the encrypted data is difficult to read.

Access control:

- Only authorized employees or applications should access backups.
- Permissions can be assigned according to user roles.
- Multi-factor authentication can provide an additional security layer.

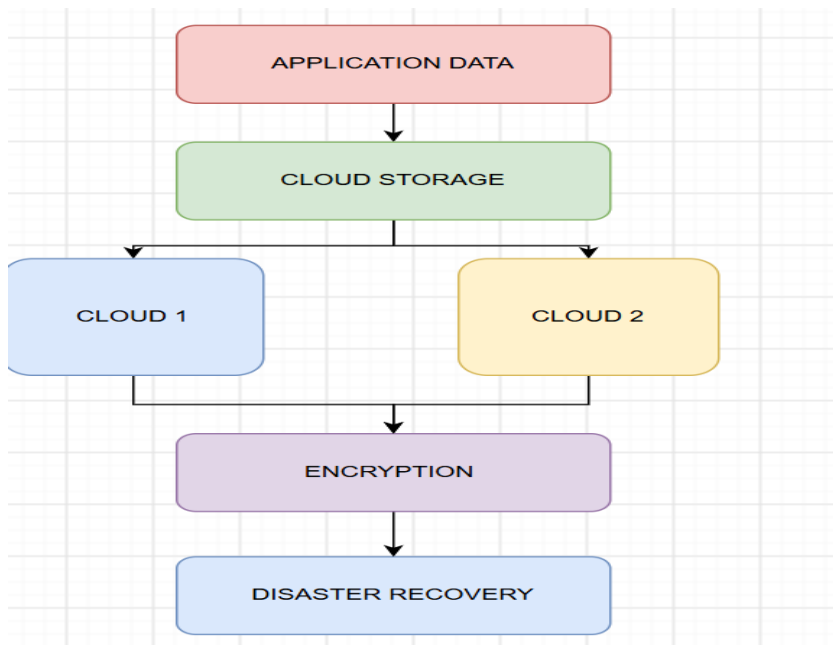
Advantages:

Compared with traditional storage:

- No need to purchase large amounts of physical hardware.
- Less spending on maintenance and data-center space.
- Storage capacity can be increased when required.
- Organizations generally pay according to their usage.
- Older backup data can be moved to cheaper archival storage.

Example:

Suppose a company's main data center is damaged due to a **fire or natural disaster**. If its backups are stored in the cloud in another location, the company can restore the required data and applications from those backups.

**Main Data Center → Cloud Backup → Disaster → Restore Data**

Cloud storage provides **redundancy, encryption, access control, and scalable storage**, making it suitable for secure backup and disaster recovery.

3.A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.

ANSWER:

A mobile application used by customers around the world may experience slow response times if all requests are handled from one central server.

How CDN improves performance

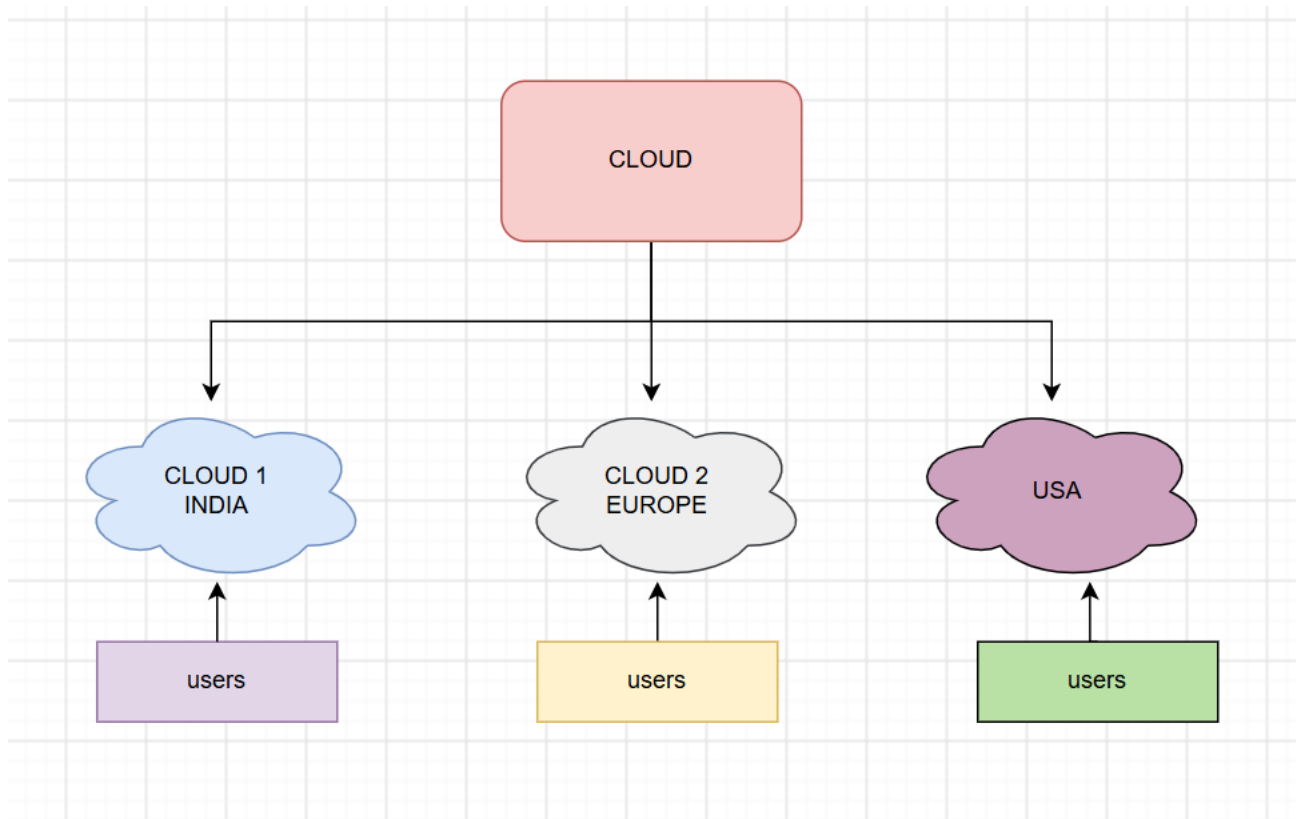
A **Content Delivery Network (CDN)** stores frequently requested content on servers located closer to users.

Instead of:

User in India → Server in USA

the request can be handled by:

User in India → Nearby Edge Server



This reduces the distance that data has to travel.

Benefits of CDN

- Reduces network latency.
- Improves loading speed.
- Handles large numbers of users.
- Reduces the workload on the main server.
- Provides faster delivery of images, videos, game files, and other content.

Role of edge servers

Edge servers are located in different geographical regions. They deliver cached content from a location closer to the user.

For example:

- Indian user → India/Singapore edge location
- European user → European edge location
- US user → US edge location

This provides a more consistent experience across regions.

Global availability through cloud providers

Cloud providers offer infrastructure in multiple geographic regions and availability zones. Applications can be deployed closer to their users and replicated across locations.

If one region has a problem, traffic can potentially be redirected to another available region.

Real-world example: Online gaming

Consider a multiplayer cloud gaming application.

Players from India, Europe, and the US should receive game assets and updates quickly. CDN edge servers can deliver these files from nearby locations, reducing download delays. Cloud infrastructure can also scale game servers according to the number of active players.

Conclusion

CDNs and edge servers bring content closer to users, **reducing latency and improving performance**, while global cloud infrastructure helps maintain availability across regions.

4.A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.

ANSWER:

A bank handles highly sensitive information such as account details, payment records, transaction histories, and customer data. Therefore, security and regulatory compliance are extremely important.

Regulatory compliance and auditing:

Cloud providers offer security and compliance features that help organizations meet industry and regulatory requirements.

They can provide:

- Compliance certifications and reports
- Audit trails
- Security policies
- Identity and access management
- Data protection mechanisms

The financial institution is still responsible for configuring and using these controls correctly.

logging and monitoring:

Logging records activities such as:

- User login attempts
- Data access
- Configuration changes
- Transaction-related activities

Monitoring helps detect:

- Unusual login behavior
- Suspicious access
- Failed requests
- Possible security incidents

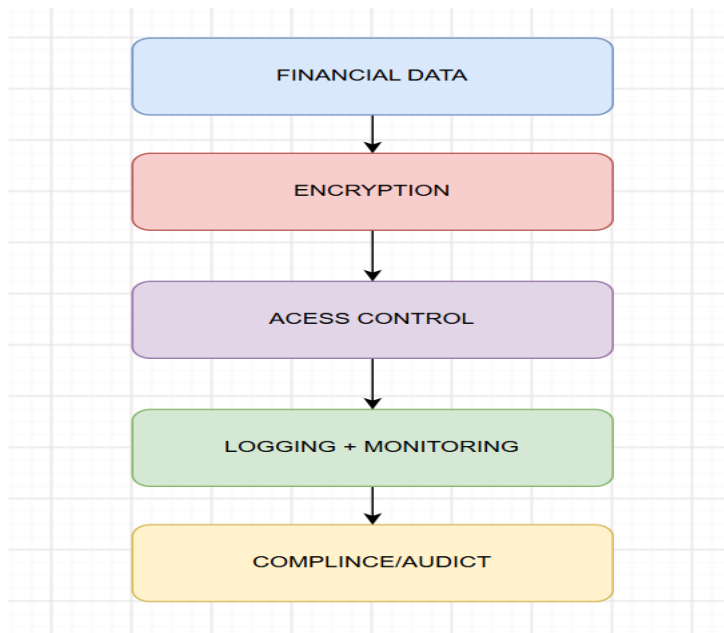
These records can also support audits and investigations.

Role of encryption

Encryption protects sensitive financial information.

For example:

Customer → Encrypted connection → Banking Cloud



Data should be protected both:

- **In transit**
- **At rest**

Encryption keys should also be properly managed and restricted.

Data sovereignty

Data sovereignty means that data may need to remain within a particular country or geographical region because of legal or regulatory requirements.

The company can:

- Select appropriate cloud regions.
- Restrict where data is stored.
- Use regional access policies.
- Monitor data movement.

Example: Online banking transaction

Suppose a customer transfers ₹50,000 through an online banking application.

The transaction data is:

1. Sent through an encrypted connection.
2. Processed by authorized services.
3. Stored securely using encryption.
4. Recorded in logs for auditing.
5. Monitored for suspicious activity.

Cloud security features such as encryption, access control, logging, monitoring, auditing, and regional data controls help financial institutions protect sensitive transactions while meeting compliance requirements.

5.A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

ANSWER:

A development team wants to deploy an application consistently across different environments such as development, testing, and production.

Containers vs Virtual Machines

Role of Kubernetes

When there are hundreds of containers, managing them manually becomes difficult. Kubernetes helps automate:

- Container deployment
- Scaling
- Service discovery
- Load balancing
- Restarting failed containers
- Updating applications

For example, if an application suddenly receives more requests, Kubernetes can increase the number of running container instances.

Portability across clouds

Containers can run consistently across:

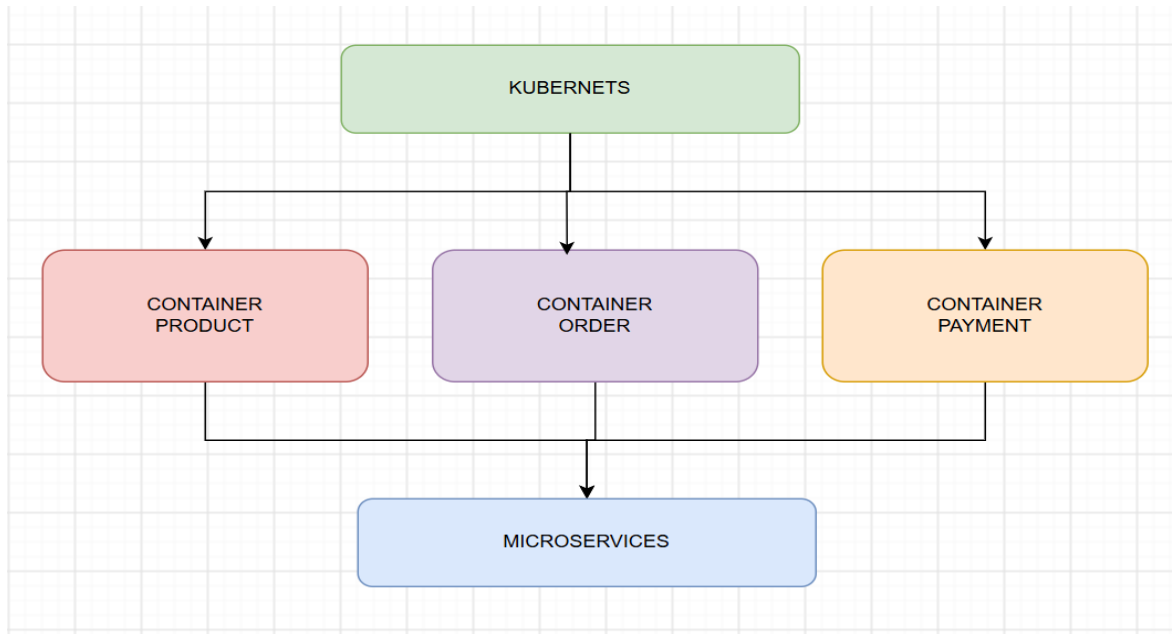
- Private cloud
- Public cloud
- Hybrid cloud
- Different cloud providers

This makes it easier to move an application without completely redesigning it for each environment.

Example:

Consider an online shopping application divided into:

User Service → Product Service → Order Service → Payment Service → Delivery Service



Each service can run inside its own container.

If the Product Service receives heavy traffic, only that service can be scaled instead of scaling the entire application.

Containers provide lightweight and portable deployment, while Kubernetes simplifies the management, scaling, and recovery of large numbers of containers. This makes containerization highly suitable for modern microservices applications.